



- (ii) The variation at position 5752 of the gene coding for a growth factor is due to a substitution mutation from **G** to **C**.

With reference to Table 2.1, describe the importance of the substitution from **G** to **C**.

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..... [3]

- (b) In a population of 268 Xinong Saanen goats:

- the frequency of the **C** allele ( $q$ ) is 0.30
- the frequency of the **G** allele ( $p$ ) is 0.70.

The Hardy-Weinberg principle can be used to predict the number of goats with **CC**, **CG** and **GG** genotypes in the population, using the equation:

$$p^2 + 2pq + q^2 = 1$$

For example, the number of goats with genotype **GG** can be predicted to be 131.

- (i) Use the Hardy-Weinberg principle to predict the number of goats with genotypes **CC** and **CG** in this population of Xinong Saanen goats.

number of goats with genotype **CC** .....

number of goats with genotype **CG** .....

[2]

Table 2.2 shows the **actual** number of goats with each genotype in a population of Xinong Saanen goats and in a population of Guanzhong goats.

**Table 2.2**

| population    | total number of goats | number of goats of each genotype |     |     | allele frequency |          |
|---------------|-----------------------|----------------------------------|-----|-----|------------------|----------|
|               |                       | CC                               | CG  | GG  | <i>p</i>         | <i>q</i> |
| Xinong Saanen | 268                   | 22                               | 116 | 130 | 0.70             | 0.30     |
| Guanzhong     | 440                   | 47                               | 69  | 324 | 0.81             | 0.19     |

- (ii) A close match between your predicted figures in (b)(i) and the actual numbers in Table 2.2 would mean that the Xinong Saanen population is in Hardy-Weinberg equilibrium.

State the name of a statistical test that could be used to find out whether or not the Xinong Saanen population is in Hardy-Weinberg equilibrium.

..... [1]

- (iii) The **predicted** numbers of goats with each genotype in the Guanzhong population according to the Hardy-Weinberg principle are:

- **CC** = 16
- **CG** = 135
- **GG** = 289.

These figures are significantly different from the actual figures in Table 2.2.

With reference to Table 2.2, describe the evidence that shows that the Guanzhong population is **not** in Hardy-Weinberg equilibrium **and** suggest reasons for this.

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..... [3]

(c) Goats can be genetically modified to produce human proteins in their milk.

In 2009, an anti-clotting protein produced in this way was approved for use as a drug in people who lack the protein.

State **one** ethical advantage and **one** ethical problem of producing medicinal drugs from the milk of genetically modified goats.

advantage .....

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problem .....

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[2]

[Total: 14]